

Aakash Rajan

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EDUCATION

Indian Institute of Technology, Madras <i>Bachelor of Technology, Mechanical Engineering</i>	CGPA: 9.03 / 10 Nov 2024 – Present
Head Start Educational Academy <i>ISC, Class XII</i>	Percentage: 96% 2024
Head Start Educational Academy <i>ICSE, Class X</i>	Percentage: 89% 2022

SCHOLASTIC ACHIEVEMENTS

- Secured an All India Rank of **2130** in JEE Advanced 2024 - Top **0.25 percentile** out of 1M candidates
- National Winner**, Caterpillar Challenge 2026 - among **7,500+** competitors nationwide
- Awarded **Best Project in Batch** for ME2400 (Mechatronics) by the former HOD, Prof. Chandramouli

RESEARCH AND TECHNICAL EXPERIENCE

Research Intern – Video World Models <i>Prof. Ravi Prakash, HiRo Lab - Indian Institute of Science (IISc)</i>	Mar 2026 – Present Bengaluru, India
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- Working on making video world models more geometry-aware for closed-loop robot navigation; modifying, training, and running inference on models including the Gaussian World Model, DINOv3, and $\pi 0.5$.
- Devised a memory-retrieval mechanism for WorldPack, improving SSIM by $\sim 3.5\%$ and LPIPS by $\sim 3\%$ over baseline on the LoopNav benchmark.
- Prototyped RoPE-based trajectory packing and geometric/semantic attention conditioning (using DINOv3) on the Navigation World Model (NWM) backbone.

Undergraduate Researcher – Multi-Agent RL for Air Traffic Control <i>Prof. Rahul Meshram, Dept. of Electrical Engineering, IIT Madras</i>	2026 – Present Chennai, India
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- Training cooperative multi-agent reinforcement learning policies for decentralized air-traffic control, as part of a challenge hosted by AI4REALNET and TU Delft.
- Aiming to improve conflict resolution and cooperation among agents under dense, dynamic airspace constraints.

Multi-Agent Autonomous Dump-Truck Simulator <i>Caterpillar National Tech Challenge</i>	Mar 2026 – Jun 2026 India
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- Led a team of 3 to develop a Python-based multi-agent dump-truck simulator for autonomous dump packing, winning Caterpillar's national contest among **7,500+** competitors with a robust, deployable solution.
- Solved the **task selection** and **optimal assignment** problem using Heuristics, the **Hungarian Algorithm**, and Monte Carlo Tree Search (MCTS).
- Integrated vehicle dynamics and path planning (**Hybrid A*-CBS**) to optimize complex truck reverse maneuvers into dump targets.

Hub Motor Cooling Jacket Design <i>Prof. Satyanarayanan Seshadri, Formula Student - Raftar Formula Racing</i>	Jul 2025 – Nov 2025 Chennai, India
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- Designed a lattice-strut based cooling jacket for an electric hub motor, optimizing pressure drop and heat-transfer coefficient backed by real life testing of LPBF manufactured cooling jacket.
- Developed a MATLAB iterative solver for cooling-loop head loss, coupled with multi-phase CFD thermal analysis to improve simulation accuracy.

Automated Medicine Dispenser <i>Prof. Sushanta Panigrahi, Mechatronics Course Project (ME2400), IIT Madras</i>	Apr 2026 – May 2026 Chennai, India
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- Designed the end-to-end pipeline, selected and integrated sensors/actuators, built CAD models, assembled the system, and programmed its functionality.
- Awarded **Best Project in Batch** by the former HOD, Prof. Chandramouli.

TECHNICAL SKILLS

Languages: Python, Java, MATLAB

Frameworks & Robotics: PyTorch, ROS2, Reinforcement Learning, World Models, AI/ML Implementation

Softwares: CAD, FEA, CFD, Simulink

RELEVANT COURSEWORK

Institute Courses		
Math and Programming	Linear Algebra	Python Data Structures & Algorithms
	Complex Networks	Differential Equations
Mechanical Engineering	Applied Thermodynamics	Fluid Mechanics
	Kinematics and Dynamics of Machinery	Basic Electrical Engineering
	Measurements Instrumentation and Control	Heat Transfer

POSITIONS OF RESPONSIBILITY

Aerodynamics and Cooling Engineer

Apr 2025 – Dec 2025

Formula Student, Raftar Formula Racing

IIT Madras

- * Part of the engineering team responsible for aerodynamic component design and manufacturing for our electric car; gained hands-on knowledge of aerodynamics, cooling strategies, vehicle dynamics & power electronics integration.
- * Looked after precise procurement, manufacturing, and assembly of all carbon fiber components.

Music Contingent Head (Piano)

2025 – Present

Inter-IIT Cultural Meet 8.0, IIT Madras

IIT Madras

- * Led a team of 8 musicians as Music Director of Stagecoach, Drama Club's highest-budget annual event.
- * Represented IIT Madras at Inter-IIT Cultural Meet 8.0 as Lead Keyboardist, placing 4th among 23 IITs.

Business Plan Presentation – Runners Up

Jun 2025 – Sept 2025

Raftar Formula Racing, Formula Bharat 2026

IIT Madras

- * Part of the team that ideated and planned an entire business plan for **Formula Bharat** 2026, securing **2nd place** out of 40+ teams across India.
- * Worked on market analysis, TCO analysis, competitive analysis, market sizing, and CAC estimation; developed a pitch deck and finance sheets for the fictional company.